

LLMs and AI Agents

Past, Present, and Future

My Journey and Insights

About Me

Wonbae Suh

Bridging customer needs !' technical implementation

About Sendbird & My Role

[Sendbird Overview](#)

[My Role as FDE](#)

Session Overview

What We'll Cover Today

Our Journey Today

Foundations of AI and LLMs

Core Principles

AI in a Nutshell

AI !' ML !' Deep Learning: Each is a subset of the previous

What is a Generative AI/LLM?

Scale and Capability

Why LLMs Are Powerful

Important Limitations

- know

Early AI: From ELIZA to Now

Past Perspective

The Evolution of AI

- Generative AI

From simple scripts to fluid, human-like dialogue

My Early Journey in LLMs

Personal Story

First Experiments

The first time I got a coherent response felt like magic

Discovering AI Agents

Combining reasoning with action is incredibly powerful

Workflows & Chain of Thought

Like assembling lego blocks of intelligence

Applied Work at SendBird

Building an AI Chatbot

SendBird AI Chatbot

Enhancing customer engagement with AI

Key Feature: Grounded in Facts

First-party data grounding was crucial for accuracy

Mitigating Hallucinations

Problem

Solutions

Lessons Learned

Seeing the system handle real support questions autonomously was surreal

SendBird AI Agent for Customer Support

Evolution from RAG Chatbot to Action-Driven Agents

From answering questions to taking actions

How LLMs Work

Technical Angle, Simplified

Training and

The Black Box and Emergence

- figured out

Both amazing and a bit scary

Why LLMs Make Errors

These limitations drive the need for augmentation techniques

Augmenting LLMs

Vectors and Memory

Retrieval-Augmented Generation (RAG)

Giving the model a smart library to consult

- open-book exam

What Are Embeddings?

- Cat
- kitty

Turning text into meaning you can measure

Vector Databases & Semantic Search

- pricing policies

Integration with LLM Prompts

This is how SendBird bot knew latest product details

Extending LLMs

Tools and Agents

Why Tools?

LLM Limitations

Tools Enable

From Static Model to Interactive Agent

- s the weather in Seoul?
- Schedule a meeting

Text + Action = Much more useful AI

Evolution: LLM Workflows vs LLM Agents

Traditional LLM Workflows

Modern LLM Agents

Native Agent SDKs

OpenAI, Anthropic, Gemini

Providers compete on making agents easier to build

Prompt Engineering vs Context Engineering

Two Approaches to Optimizing LLM Interactions

Prompt Engineering

Context Engineering

- frame

Production AI systems need both: clear prompts AND well-engineered context

Model Context Protocol (MCP)

Introduced by Anthropic, late 2024

Like HTTP for AI-tool communication

Agent-to-Agent Protocol (A2A)

Announced by Google, 2025

Building networks of collaborative AI

Core Components of AI Agents

Building Blocks of Autonomous Systems

Each component works together to create intelligent, autonomous behavior

Agentic Compound Systems

We are at the dawn of AI ecosystems

LLMs in Education

Impact on Learning and Teaching

Student Adoption

Two in three students now use ChatGPT

The Double-Edged Sword

Concerns

Opportunities

Teachers' Perspectives

Pew Survey, late 2023

The Path Forward: AI Literacy

AI isn't going away – manage it thoughtfully

LLMs in the Workplace

Changing How We Work

Widespread Adoption

Each knowledge worker now has a junior assistant

Productivity Boosts

MIT Study, 2023

New Workflows and Roles

- prompt engineer

Augmenting roles rather than replacing them

Workplace Challenges

Concerns

Solutions

Overall Impact

Professionals who combine expertise with AI will be in high demand

What Lies Ahead

The Future of LLMs and Generative AI

Even More Capable Models

Multimodal and Versatile AI

Not confined to chat boxes anymore

Agent Ecosystems

- AI CEO

Managing fleets of AI workers

Ethics, Regulation, and Society

AI for Societal Challenges

AI should amplify human potential, not undermine it

"The illiterate of the 21st century will not be those who cannot read and write, but those who cannot learn, unlearn, and relearn."

— Alvin Toffler

Continuous Learning for Humans

Adaptability is the superpower of the 21st century

Key Takeaways and Advice

Lessons for the Journey Ahead

1. Stick to Fundamentals

2. Learn Step by Step

3. Hands-On Experimentation

Get your hands dirty

4. Stay Curious and Keep Learning

5. Be Ready to Unlearn/Adapt

6. Ethics and Purpose

Remember the why

Final Thoughts

"The AI revolution is not just something happening to us; it's something we are all co-creating."

Thank You!

Questions & Discussion

Let's explore the future of AI together